

Class 9: Structural Bioinformatics Part 1

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The PDB database

Download a CSV file from the PDB site (accessible from “Analyze” > “PDB Statistics” > “by Experimental Method and Molecular Type”. Move this CSV file into your RStudio project and use it to answer the following questions:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	176,204	20,299	12,708	342	218
2	Protein/Oligosaccharide	10,279	3,385	34	8	11
3	Protein/NA	9,007	5,897	287	24	7
4	Nucleic acid (only)	3,066	200	1,553	2	15
5	Other	173	13	33	3	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	83 32	209,886				
2	1 0	13,718				
3	0 0	15,222				
4	3 1	4,840				
5	0 0	222				
6	0 4	22				

```
stats$Total
```

```
[1] "209,886" "13,718" "15,222" "4,840" "222" "22"
```

There are not numeric...

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
stats
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM    NMR Integrative `Multiple methods` Neutron
  <chr>              <dbl> <dbl> <dbl>      <dbl>          <dbl>  <dbl>
1 Protein (only)      176204 20299 12708      342            218    83
2 Protein/Oligosacch~  10279  3385   34         8             11     1
3 Protein/NA          9007  5897  287        24             7     0
4 Nucleic acid (only)  3066   200  1553        2            15     3
5 Other               173    13   33         3             0     0
6 Oligosaccharide (o~   11     0    6         0             1     0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.total <- sum(stats$Total)
n.total
```

```
[1] 243910
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy. Give your answer to 2 significant figures.

```
n.xray <- sum(stats$`X-ray`)
xray_structures <- (n.xray / n.total)*100
round(xray_structures, 2)
```

```
[1] 81.48
```

81.48% are X-ray

```
n.em <- sum(stats$EM)
em_structures <- (n.em / n.total)*100
round(em_structures, 2)
```

```
[1] 12.22
```

12.22% are EM

Q2: What proportion of structures in the PDB are protein?

```
n.protein <- sum(stats$Total[1:3])
prot_structures <- (n.protein/n.total)*100
round(prot_structures, 2)
```

```
[1] 97.92
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,150 HIV-1 protease structures in the current PDB.

Exploring PDB structures

Package for structural bioinformatics

```
library(bio3d)
hiv <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hiv
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Let's first use the Mol* viewer to explore this structure.

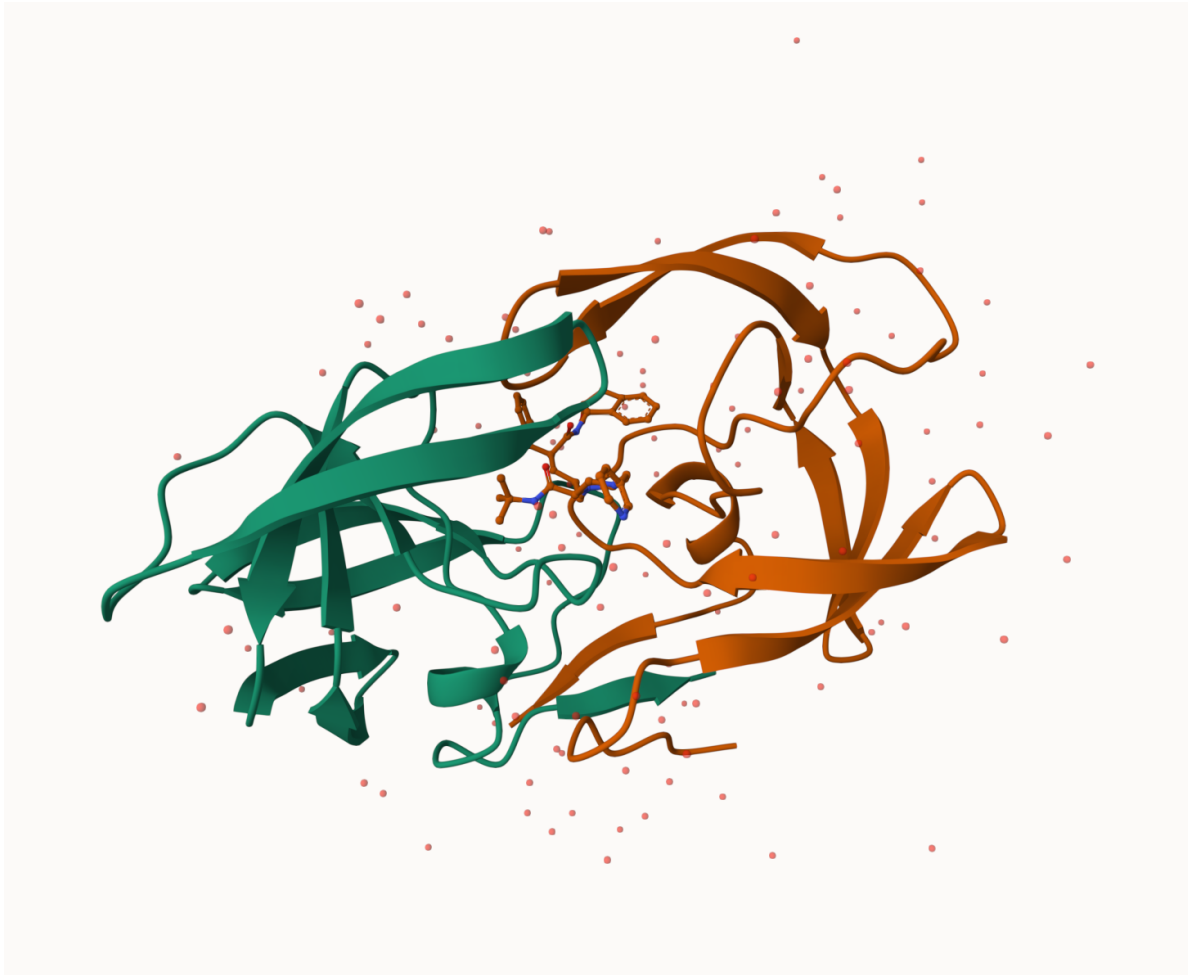


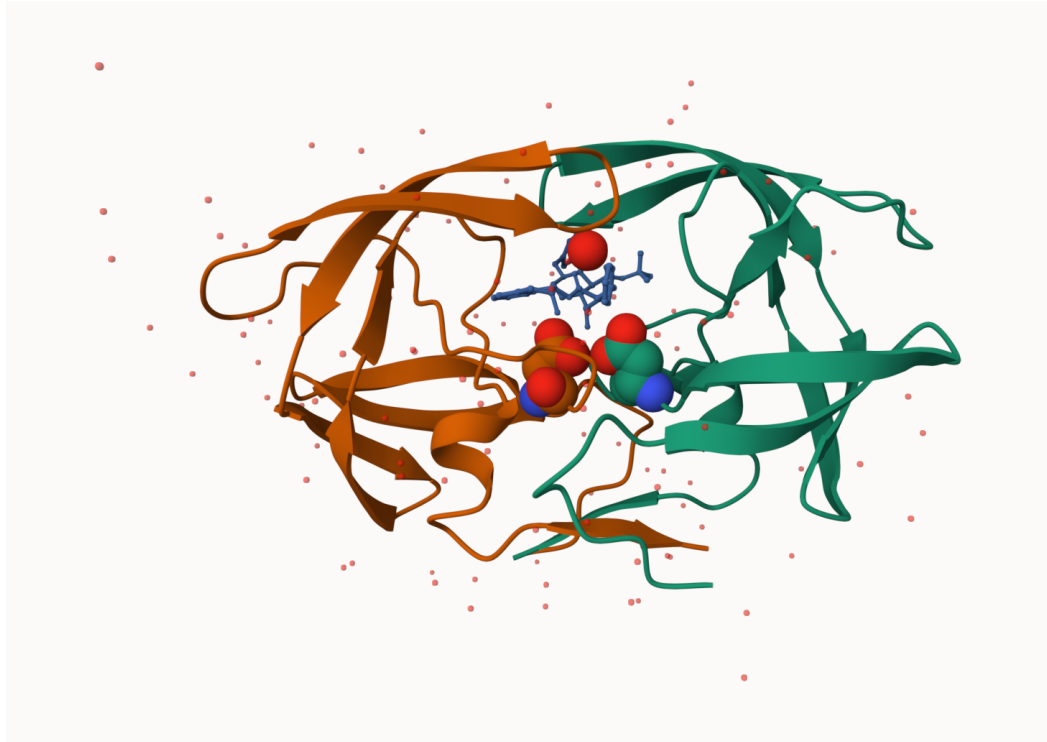
Figure 1: My first view of HIV-Pr

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We only see one atom because hydrogens are omitted to avoid clustering and improve visibility. Also, in real models hydrogens are difficult to see and including them would not accurately describe the reality. We assume they are there.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

HOH 308



Q6:

PDB objects in R

Extract the sequence

`pdbseq(hiv)`

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
"P"	"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
"E"	"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
"R"	"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
"Q"	"I"	"L"	"I"	"E"	"I"	"C"	"G"	"H"	"K"	"A"	"I"	"G"	"T"	"V"	"L"	"V"	"G"	"P"	"T"
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	1
"P"	"V"	"N"	"I"	"I"	"G"	"R"	"N"	"L"	"L"	"T"	"Q"	"I"	"G"	"C"	"T"	"L"	"N"	"F"	"P"
2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"	"E"
22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"	"R"
42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61
"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"	"Q"

```

62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"

```

```
chainA_seq <- pdbseq(trim.pdb(hiv, chain = "A"))
```

I can interactively view these PDB objects in R with the new **bio3dview** package.

```
library(bio3dview)
#view.pdb(hiv)
```

```
library(bio3dview)
#sel <- atom.select(hiv, resno = 25)
#view.pdb(hiv, highlight = sel, highlight.style = "spacefill", colorScheme = "chain", col = c
```

Predict protein flexibility

We can run a bioinformatics calculation to predict protein dynamics - i.e. functional motions.

```
adk <- read.pdb("6s36")
```

```
Note: Accessing on-line PDB file
      PDB has ALT records, taking A only, rm.alt=TRUE
```

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```

Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

```

Protein sequence:

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRITIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 214 amino acid residues in this pdb object.

Q8: Name one of the two non-protein residues?

In this case there are more than two but one of them is CL.

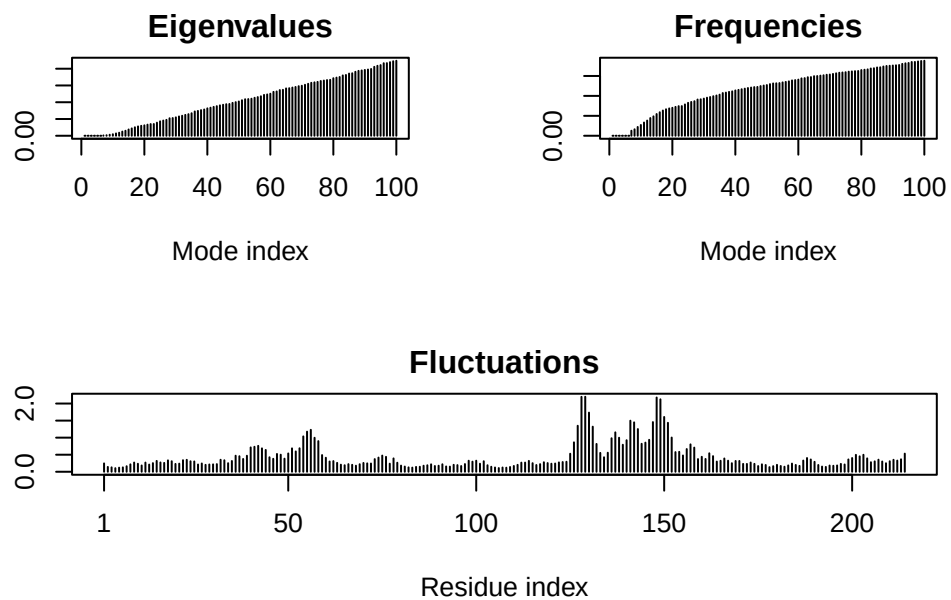
Q9: How many protein chains are in this structure?

There is only one protein chain.

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.02 seconds.  
Diagonalizing Hessian... Done in 0.33 seconds.
```

```
plot(m)
```



Generate a “trajectory” of predicted motion

```
mktrj(m, file = "ADK_nma.pdb")
```

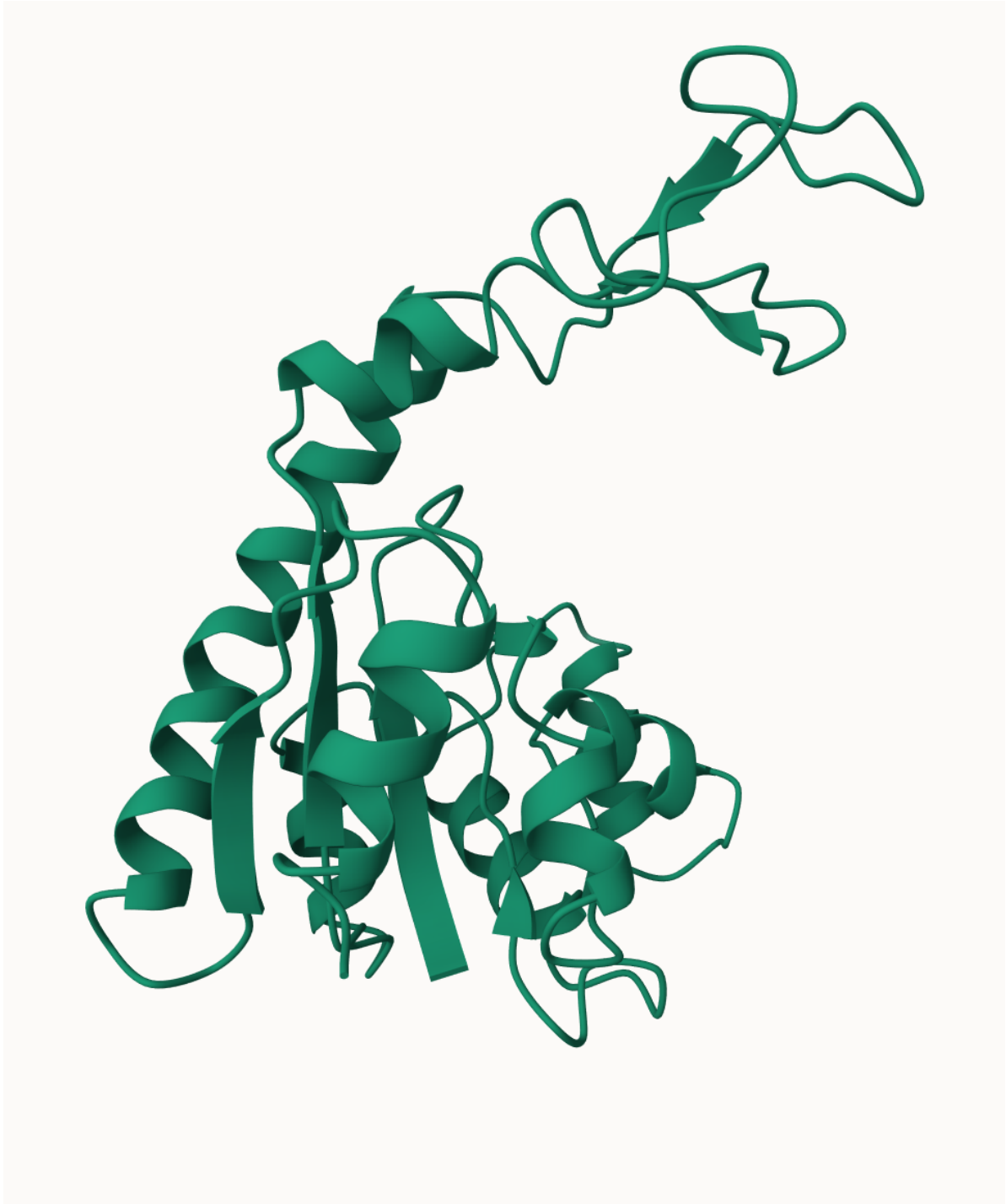



Figure 2: ADK prediction